technische universität dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Qualitative Process Analysis

Christian Janiesch | Business Process Management | Dortmund 2025-06-05

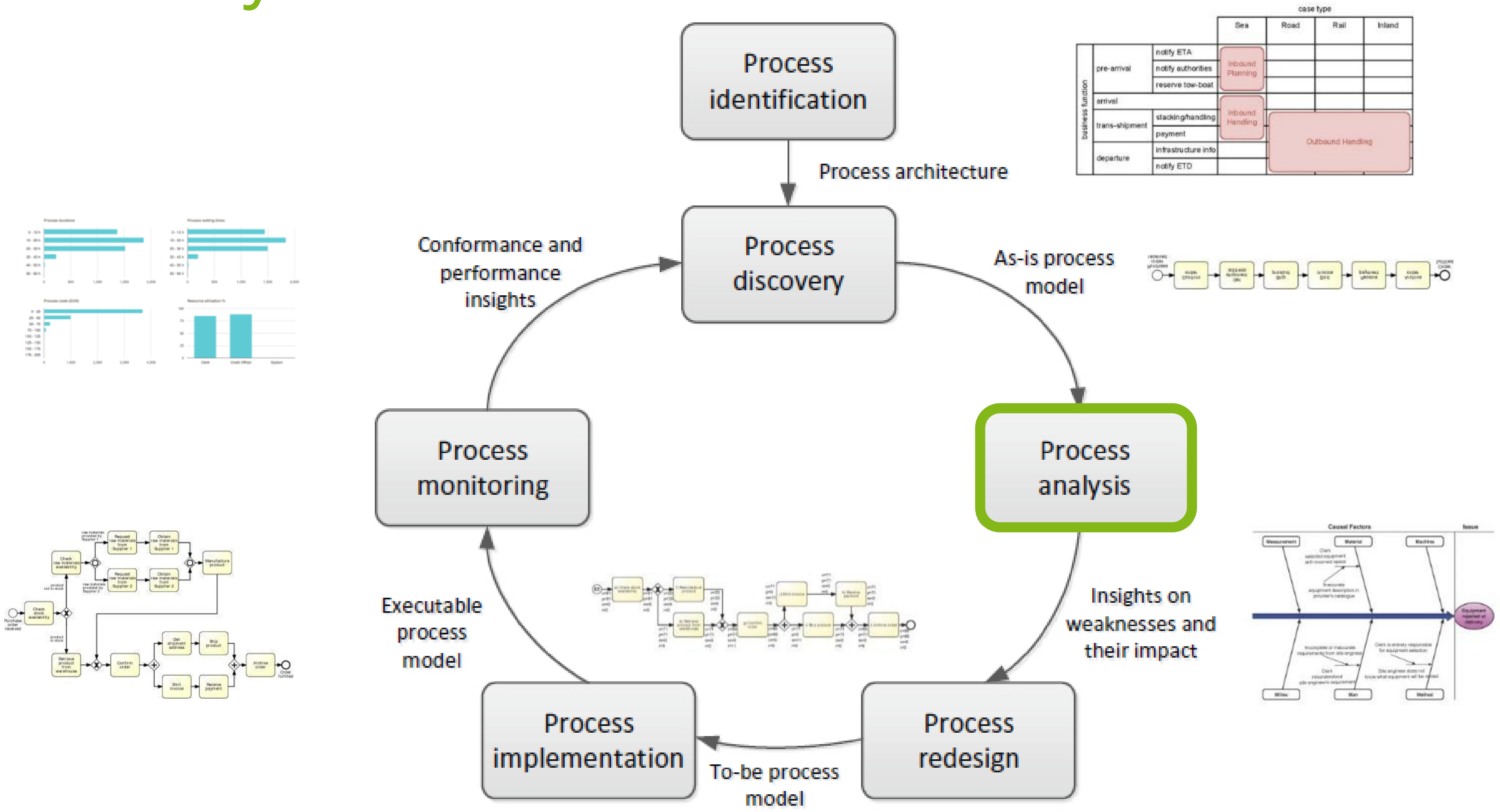
Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

- Understand and be able to use value-added analysis
- Understand and be able to use waste analysis
- Understand and be able to use stakeholder analysis and issue documentation
- Understand and be able to use root cause analysis

The BPM Lifecycle



Dumas et al. (2018)

Process Analysis

“Quality is free, but only to those who are willing to pay heavily for it.”

Tom DeMarco (1940–)

- Analyzing Processes is an **art** and a **science**
- **Qualitative analyses** use a range of **principles and techniques**
- **Quantitative analyses** use **measures** to judge a process
- There is no single way of producing good results

Dumas et al. (2018)

A Selection of Process Analysis Techniques

- **Qualitative** analysis
  - Value-added Analysis
  - Waste Analysis
  - Stakeholder Analysis
  - Root cause Analysis

- **Quantitative** Analysis
  - Flow Analysis
  - Queueing Analysis
  - Process Simulation

Dumas et al. (2018)

Purpose of Qualitative Process Analysis

- Identify and eliminate **unnecessary steps in a process**
  - Valued-added analysis
  - Waste analysis

- Identify, document, understand, and prioritize **issues**
  - Stakeholder analysis and issue documentation
  - Root cause analysis
  - Pareto charts, PICK charts

Dumas et al. (2018)

Value-added Analysis

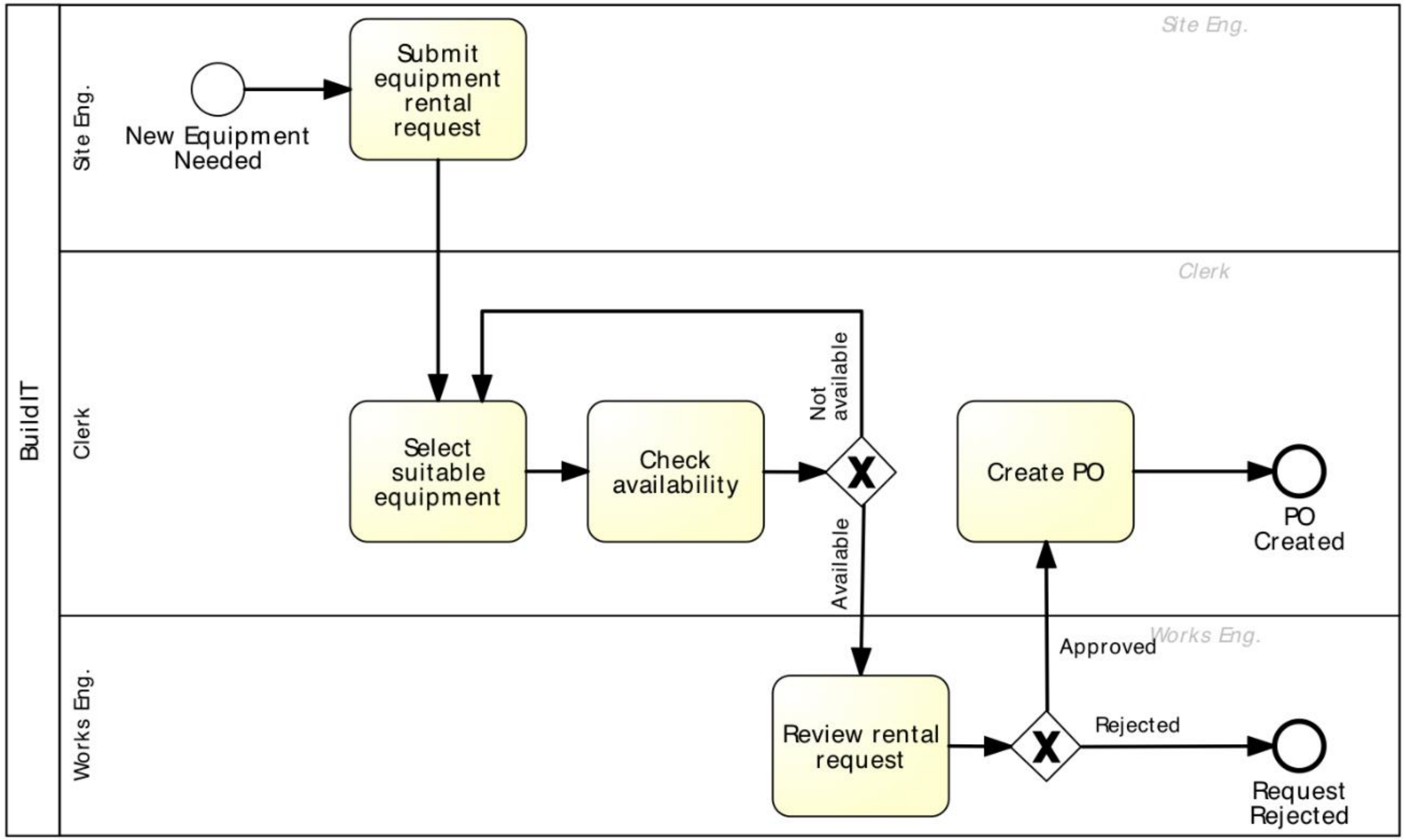
   

Value-added Analysis

- Technique to **identify unnecessary steps** in a process and eliminate them
- **Decompose** the process into steps and identify customer & positive outcomes
  - Analyze documentation
    - e.g., process models, checklists
  - Analyze tacit knowledge
    - e.g., through observation and interviews
- **Classify** each step into
  - **Value-adding (VA):** Produces value or satisfaction to the customer
  - **Business value-adding (BVA):** Necessary or useful for the business to run smoothly, or required due to the regulatory environment, e.g. checks, controls
  - **Non-value-adding (NVA):** Everything else including handovers, delays, and rework

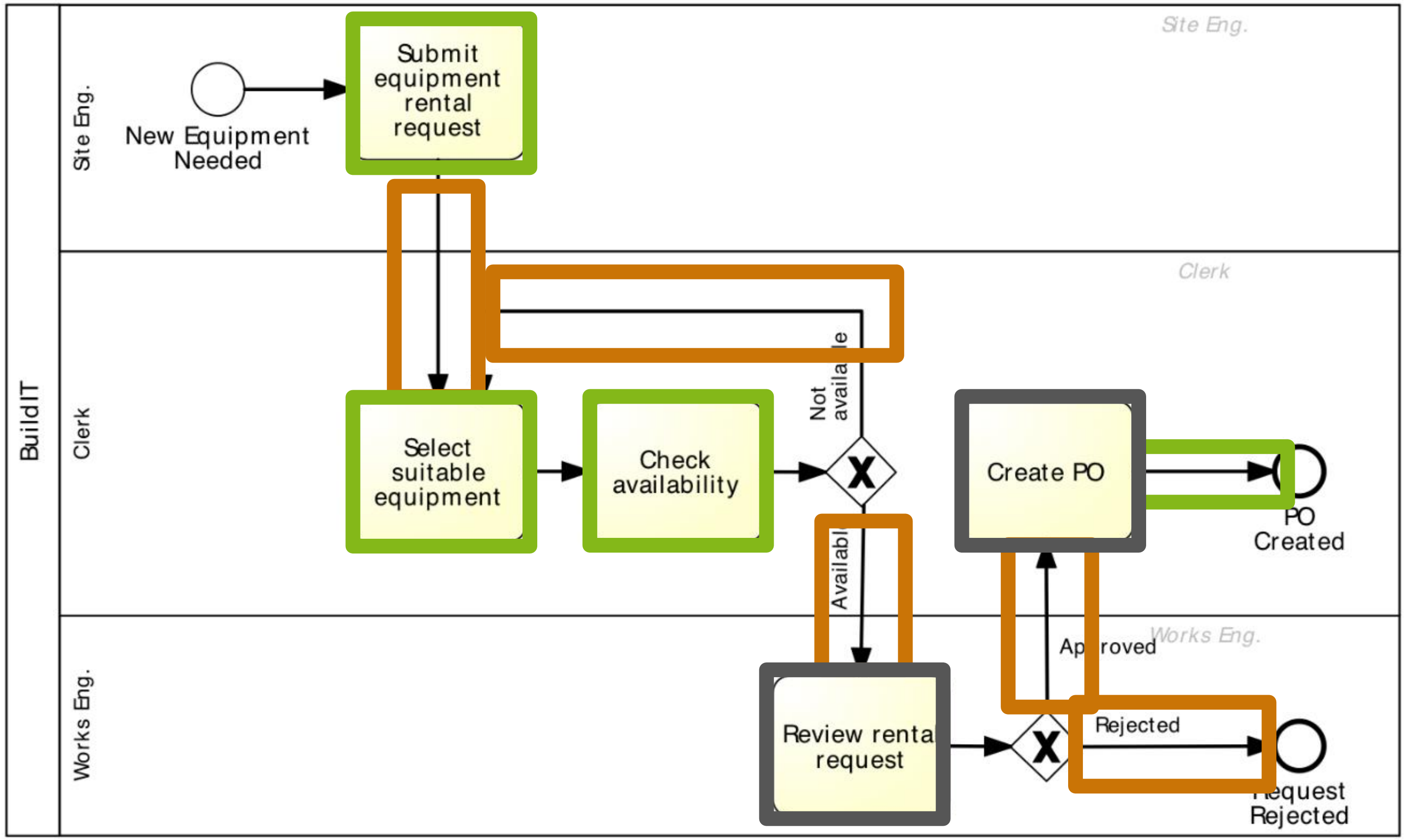
Dumas et al. (2018)

Value-added Analysis Example



Dumas et al. (2018)

Value-added Analysis Example



Dumas et al. (2018)

Guidelines for Value-added Analysis

- **Handoff steps** between process participants are **non-value adding**
- **Control steps** are generally **business-value adding** (an excessive amount may be NVA)
- Creating a **formal document** in general is typically **non-value adding**
- **Steps** imposed by **accounting or legal** requirements are generally **business-value adding** (beware of hearsay)
- Differentiate between **passing** information (NVA) and **creating** and **making** information **available** (VA)
- **Classification is to some extent subjective and depends on context!**
(That is why it is an art!)

Dumas et al. (2018)

Guidelines for Elimination

- **Minimize and Eliminate NVA steps**
  - Use automation (BPMS)
  - Reconsider participants (e.g., clerk)
  - Consider process variants (e.g., low value cases without approval)

- **Minimizing and Eliminating BVA steps is a trade-off**
  - Consider the value for the company w.r.t. goals and strategy
  - Consider regulations

- More about this when we discuss process redesign

Dumas et al. (2018)

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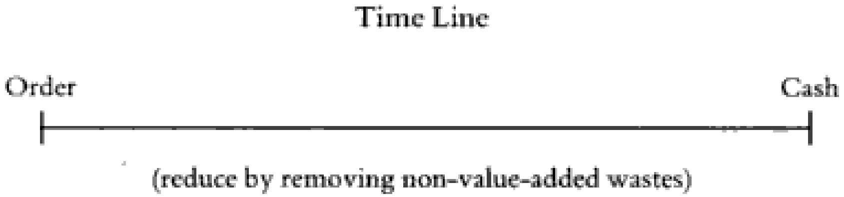
Waste Analysis

Waste Analysis

*"All we are doing is looking at the timeline,
from the moment the customer gives us an order
to the point when we collect the cash.*

*And we are reducing that timeline
by removing the non-value-added wastes."*

Taiichi Ōhno (1912–1990)



Ohno (1988)

7+1 Sources of Waste

- **Transportation** (send, receive)
- **Motion** (drop-off, pick-up, go to)
- **Inventory** (large work-in-progress)
- **Waiting** (idle time between tasks)
- **Defects** (compensation to fix defects)
- **Overprocessing** (performing what is not needed)
- **Overproduction** (unnecessary cases)

- **Skills** (resource underutilization, idle resources)

- **Move**
  - Transportation & motion
- **Hold**
  - Inventory (+ *overprovisioning*) & waiting
- **Overdo**
  - Defects, overprocessing, overproduction

Ohno (1988), Liker (2004), Six Sigma, Dumas et al. (2018)

Move

- **Transportation**
  - Physical transport of materials from A to B
  - Physical transport of documents
  - Handoffs between participants
  - → Electronic Data Interchange (EDI) has automated some, lanes in pools help to identify further waste
- **Motion**
  - Physical movement of process participants from A to B
  - Application switch in digitalized processes
  - → Robotic Process Automation (RPA) can help to reduce some of this waste

Dumas et al. (2018)

Hold

- **Inventory**
  - Physical hold of more stock than necessary to perform a task
  - Work-in-process
  - *(Resource/skill overprovisioning – more capacity than necessary)*

  → See Quantitative Process Analysis
- **Waiting**
  - Change-over times
  - Idle times, suspend times
  - Resource availability
  - **Transportation is a constant source of waiting waste**

  → See Quantitative Process Analysis

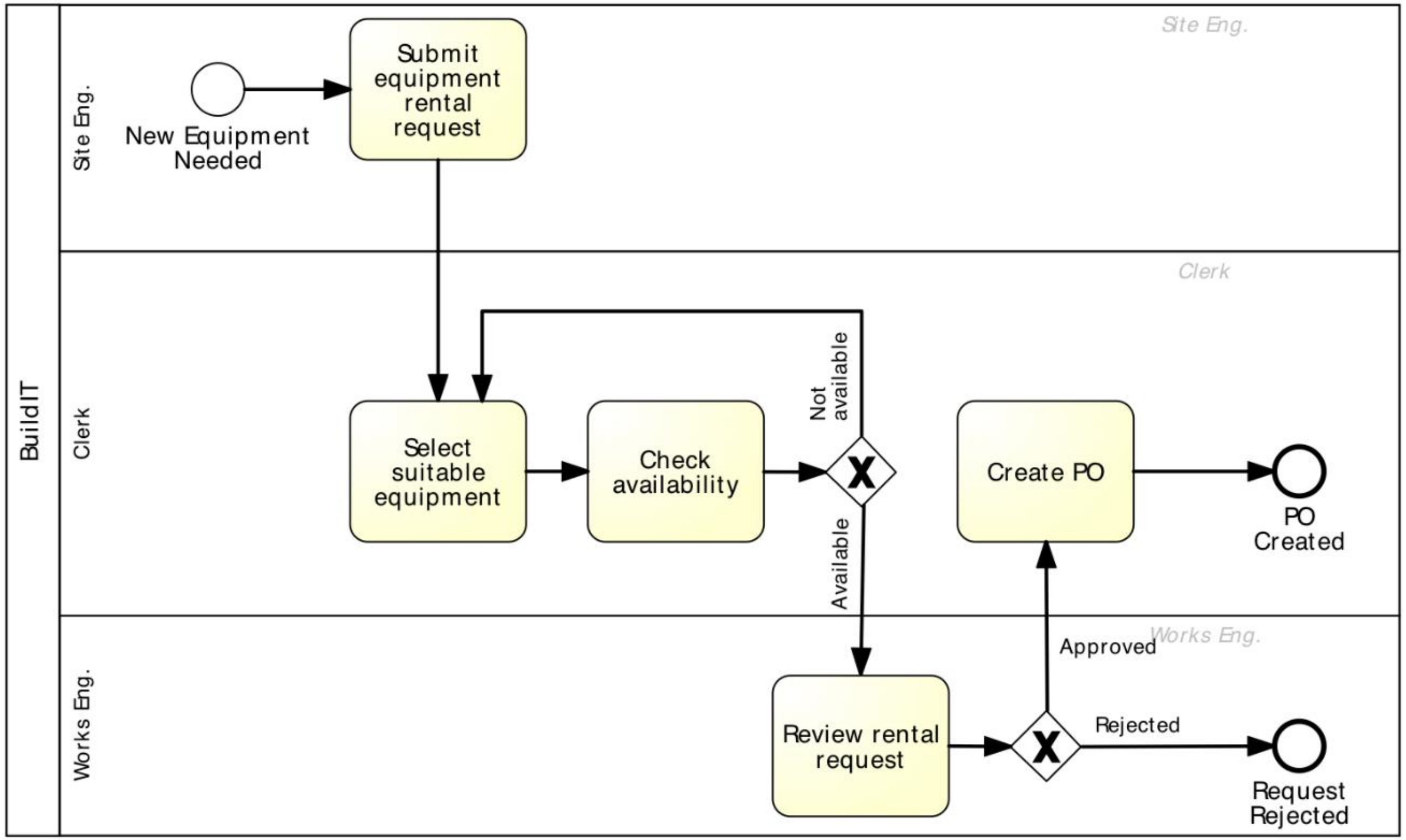
Dumas et al. (2018)

Overdo

- **Defect**
  - Correct, repair, or compensate for issues
  - Results in rework (loops)
  - → See Process Redesign
- **Overprocessing**
  - Unnecessary tasks or unnecessary perfectionism
  - → See Process Redesign
- **Overproduction**
  - Successful process instances that were not necessary (e.g., unrequired approval)
  - Failed process instance that should not have been initialized (e.g., invalid application)
  - → See Process Redesign

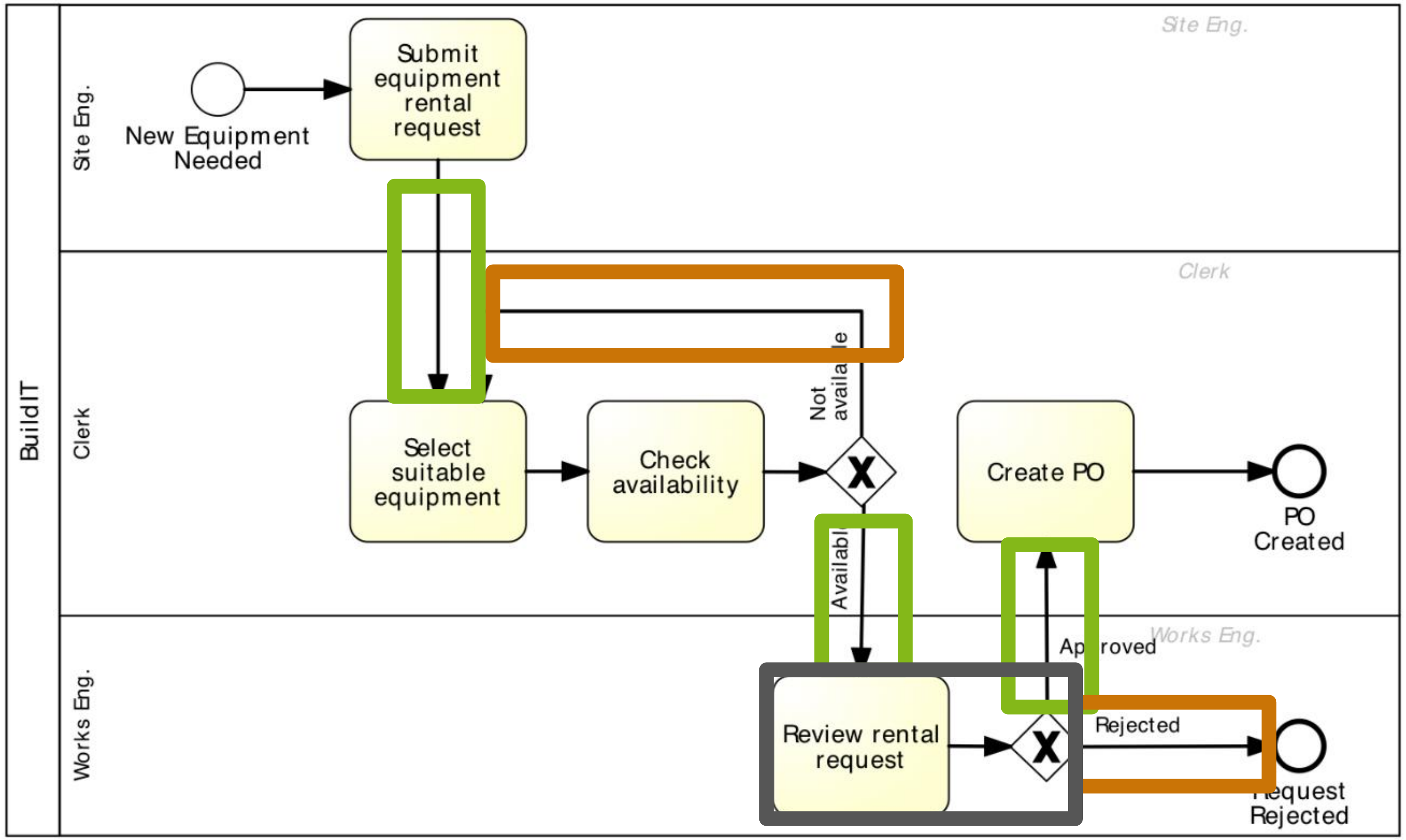
Dumas et al. (2018)

Value-added Analysis Example



Dumas et al. (2018)

Value-added Analysis Example



Dumas et al. (2018)

Stakeholder Analysis & Issue Documentation




Stakeholder Analysis

- Gathering data from **multiple sources** by interviewing stakeholders of different types and reconciling their viewpoints
  → See Process Discovery
- There are typically **five categories of stakeholders**
  - **Customers** of the process
  - **Process participants**
  - **Process owner** and operational managers who supervise the process participants
  - **External parties**, e.g., suppliers, sub-contractors
  - **Sponsor** of the process improvement effort and other executive managers who have a stake in the performance of the process

Dumas et al. (2018)

Stakeholder Concerns

- **Customers**
  - Slow cycle time, defects, lack of transparency, or lack of traceability (inability to observe the current process status)
- **Process participants**
  - Resource utilization, working under stress, defects arising from handoffs and wastes
- **Process owner**
  - Performance (cycle times, processing times, common defects and wastes, compliance)
- **External parties**
  - Steady or growing stream of work from the process, ability to plan ahead, meet SLA
- **Sponsor**
  - Strategic alignment of the process, contribution to key performance indicators

Dumas et al. (2018)

Issue Documentation

- **Issue register**
  - Categorize identified **issues and their impact** as part of as-is process modelling
  - Can contains **issues** and **factors**

Name	Description	Qualitative Impact	Assumptions	Quantitative Impact
Rejected equipment	Site engineers reject delivered equipment due to non-conformance to their specifications	Disruption to schedules. Employee stress and frustration	• We rent 3000 pieces of equipment p.a. • Each time an equipment is rejected due to an internal mistake, we are billed the cost of one day of rental, that is 100 €. • 5% of them are rejected due to an internal mistake	3.000 × 0.05 × 100 € = **15.000** € p.a.

Dumas et al. (2018)

Issue Documentation

- **Issue register tables** can have the following columns
  - Issue number
  - Issue name
  - Description/ explanation
  - Priority
  - Assumptions/ input data
  - Impact: qualitative (cost) and quantitative (nuisance)
  - Caused by/ causes
  - Possible solution

Dumas et al. (2018)

Impact Assessment

- A small **subset of issues** in the issue register are likely responsible for the **largest share of impact**
- For a given issue, a small **subset of factors** behind this issue are likely responsible for the **largest share of occurrences** of this issue
- Techniques necessary to **prioritize** a collection of issues or factors behind an issue
  - Pareto analysis: 80-20 principle
  - Growth-share matrix
  - ...

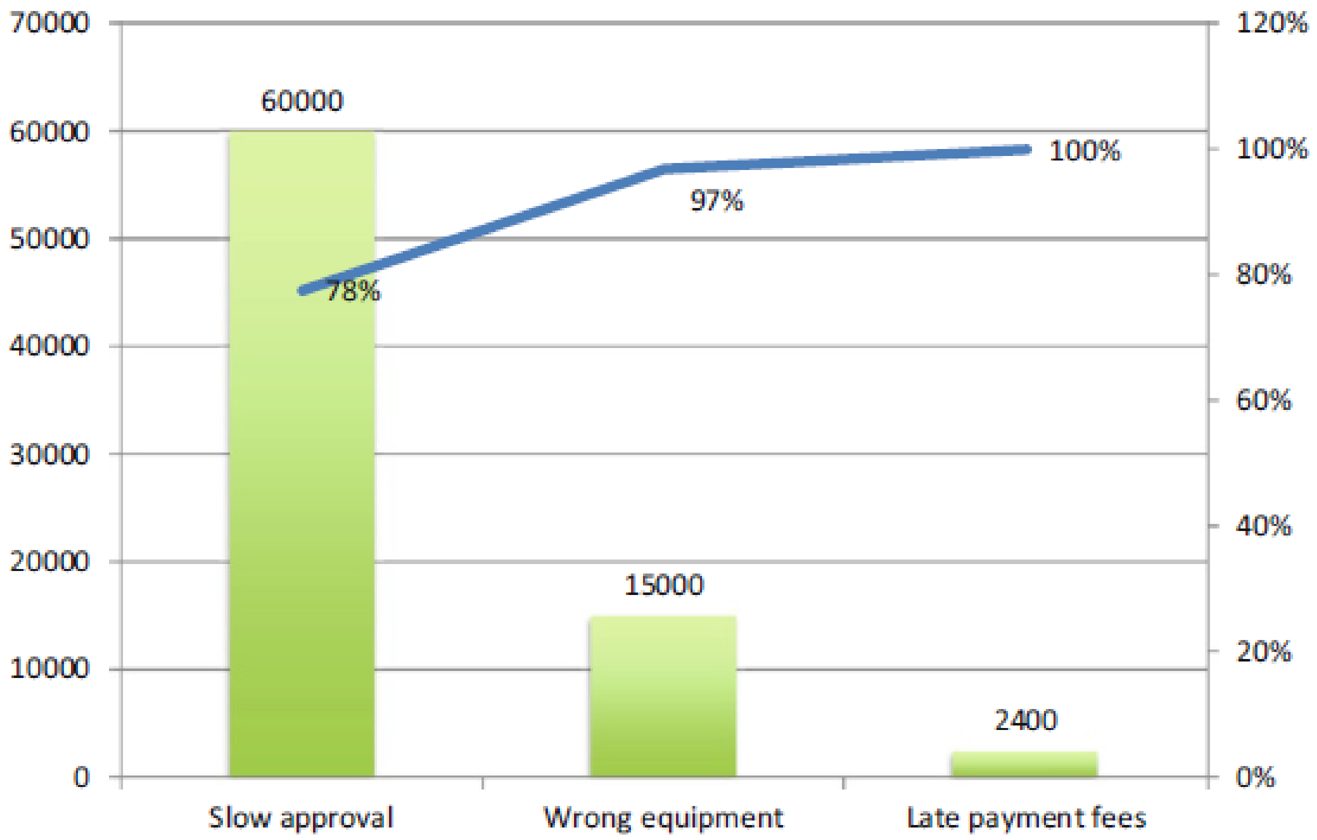
Dumas et al. (2018)

Impact Assessment Process – Pareto Analysis

- Few problems are responsible for large parts of the deficiencies
- **Define** the **effect** to be analyzed and the **measure** via which this effect will be quantified
- Identify all **relevant issues** that contribute to the effect to be analyzed
- **Quantify each issue** according to the chosen measure
  - Can be done on the basis of the issue register, i.e. the quantitative impact column
- Sort the issues according to the chosen measure (from highest to lowest impact) and draw a so-called **Pareto chart**

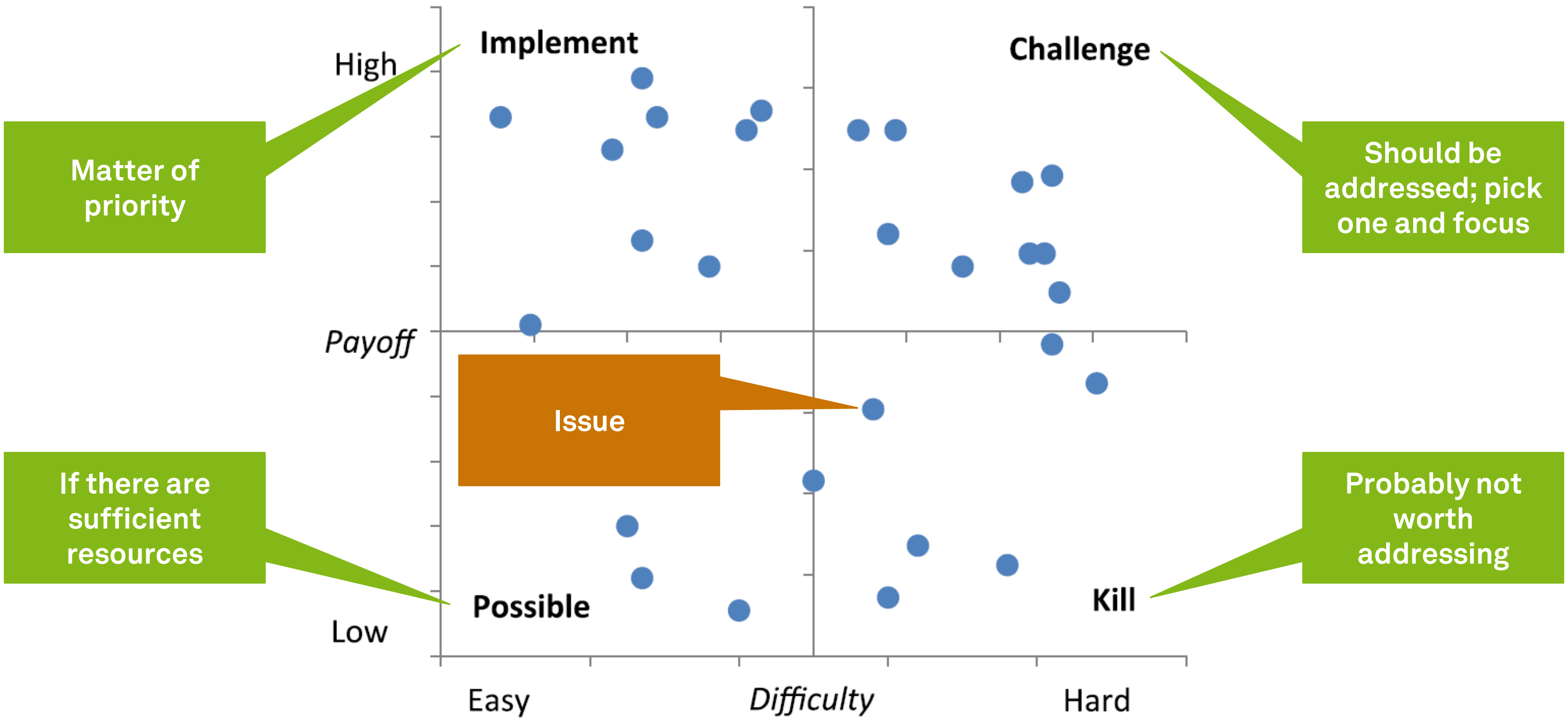
Dumas et al. (2018)

Pareto Chart



Dumas et al. (2018)

PICK Chart



Dumas et al. (2018)

Root Cause Analysis



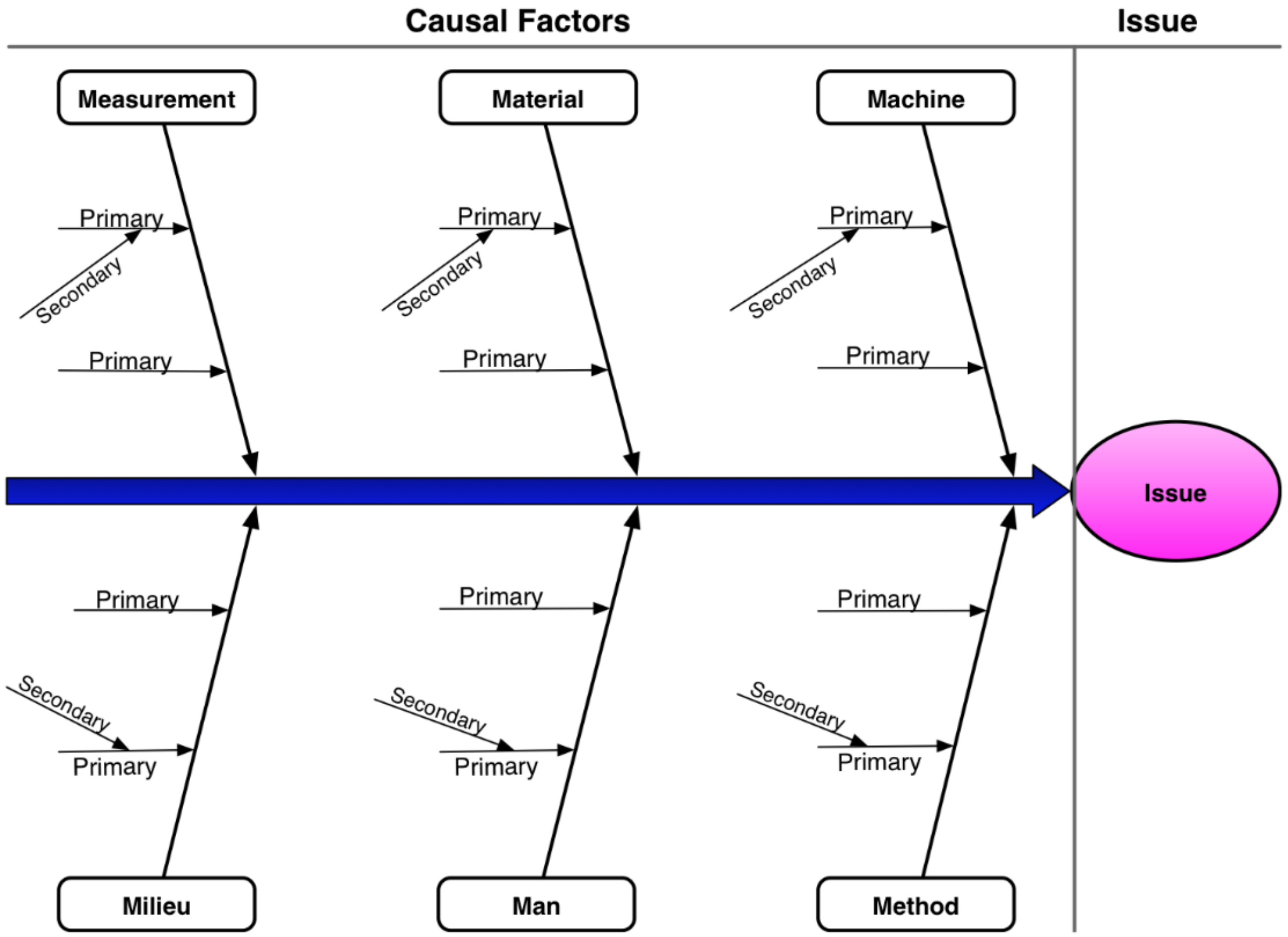

Root Cause Analysis

- Root cause analysis is a **family of techniques**
- Goal is to **understand root causes** of problems why a process does not have a better performance
- Distinguishes
  - **Causal factors** (root cause)
  - **Contributing factors** (enabler, multiplier)

- Analysis are usually performed as **interviews** or **workshops**
  - Be aware of multiple perspectives/ perceptions on one issue
- Results are documented as
  - **Cause-Effect diagrams**
  - **Why-Why diagrams**

Dumas et al. (2018)

Cause-Effect Diagram



Dumas et al. (2018)

The 6 M (1/3)

- **Machine** (technology)
  - **Lack of functionality** in application systems
  - **Redundant storage** of data across systems, leading e.g. to double data entry and data inconsistencies
  - **Low performance** of IT of network systems, leading e.g. to low response times for process participants
  - **Poor user interface** design, leading e.g. to process participants not realizing that some data are missing or available
  - **Lack of integration** between multiple systems within the enterprise or with external systems

Dumas et al. (2018)

The 6 M (2/3)

- **Method** (process)
  - **Unclear, unsuitable or inconsistent assignment** of decision-making and processing responsibilities to process participants
  - **Lack of empowerment** of process participants, leading to process participants not being able to make decisions
  - **Excessive empowerment** may lead to process participants having too much discretion and causing losses to the business
  - **Lack of** timely **communication** between process participants
- **Material**
  - **Incorrect** data leading to a wrong decision being made during the execution of a process

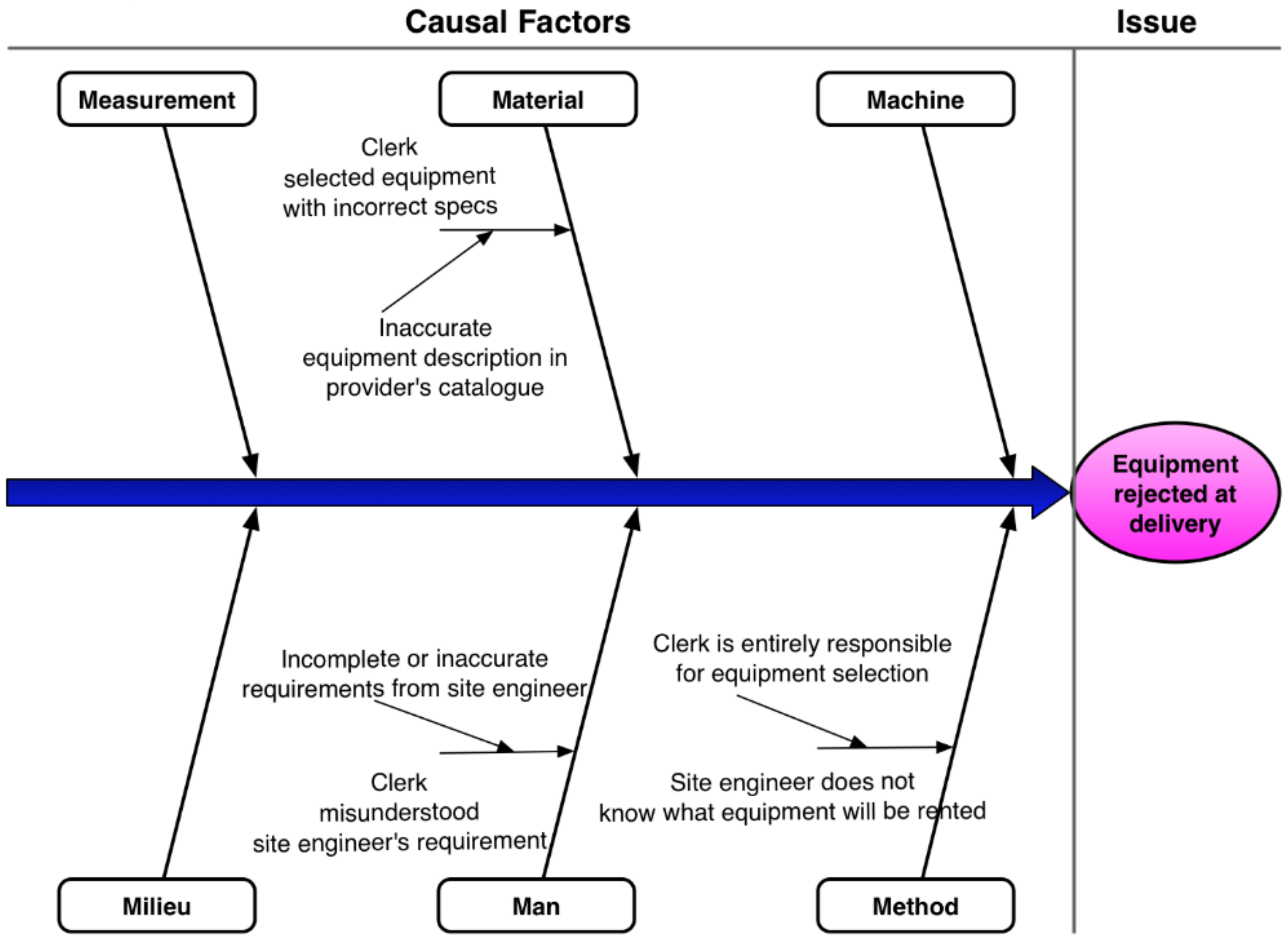
Dumas et al. (2018)

The 6 M (3/3)

- **Man**
  - **Lack of training** and clear instructions for process participants
  - **Lack of incentive system** to motivate process participants sufficiently
  - **Expecting** too much from process participants
  - **Inadequate recruitment** of process participants
- **Measurement**
  - **Mistakes in measurements** or calculations made during the process
- **Milieu**
  - Factors stemming from the **environment** in which the process is executed (customer, suppliers or other external actors)
  - Distinction between global, external, and internal factors

Dumas et al. (2018)

Cause-Effect Diagram – Example



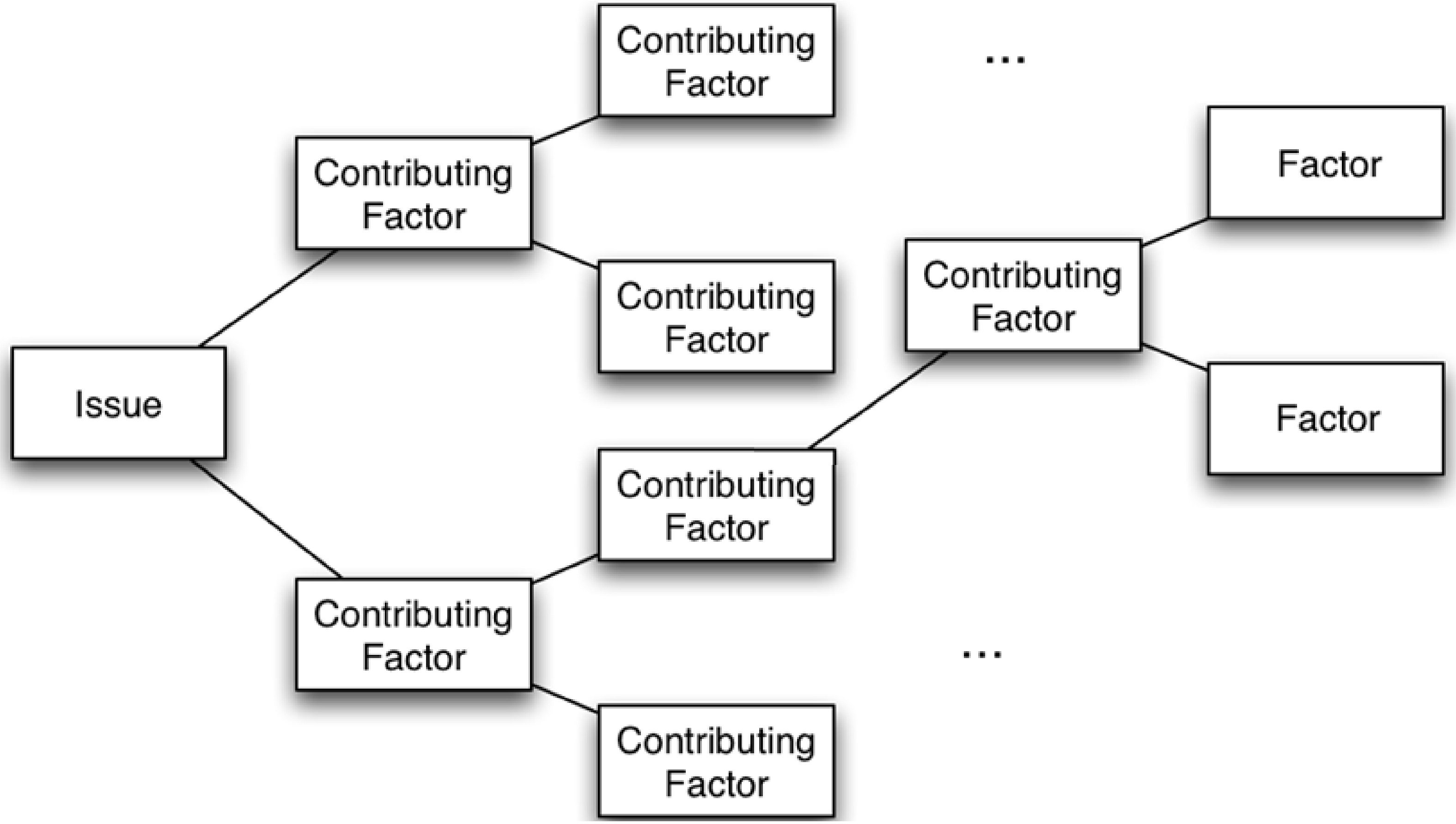
Dumas et al. (2018)

Why-Why Questions

- Recursively ask the question **"Why?"**
  - "Why does this issue happen?"
  - "What is/are the main sub-issue(s) leading to this issue?"
- After asking **five times** you should arrive at the root-cause
- At level 3 to 4 on should focus on **issues that can be solved**
- **Leaves** should be **fundamental** factors (atomic, cannot be explained otherwise)

Dumas et al. (2018)

Why-Why Diagram



Dumas et al. (2013)

Why-Why Diagram - Example

- **Issue** "Site engineers sometimes reject delivered equipment"
  why?
  - "Wrong equipment is delivered"
    why?
    - "equipment descriptions in supplier's catalog not accurate"
    - "miscommunication between site engineer and clerk"
      why?
      - "site engineer provides only brief/ inaccurate description of what they want"
      - "site engineer does not (always) see the supplier catalogs when making a request and does not communicate with the supplier"
        why?
        - » site engineer generally **does not have Internet connectivity**

Dumas et al. (2018)

Qualitative Process Analysis

- **This is not the end**

- There are tons of qualitative techniques which could be used for process analysis
  - Six Sigma techniques
  - Causal factor charts (capture context)
  - Theory of constraints (identify bottlenecks)
  - ...

Dumas et al. (2013)

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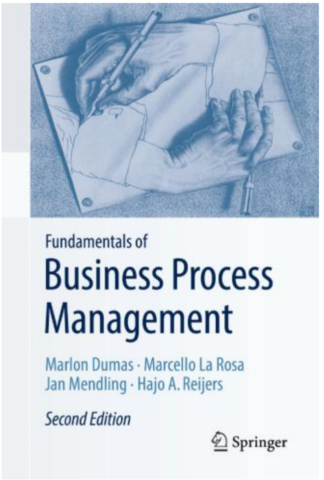
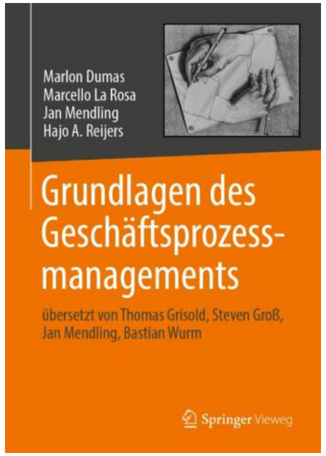
Recap Questions

- Why is process analysis an art and a science?
- What are different techniques of qualitative process analysis and what are their key benefits?
- Can you give an own example for the application of each qualitative process analysis technique?

Literature

- **Section 6:** Qualitative Process Analysis





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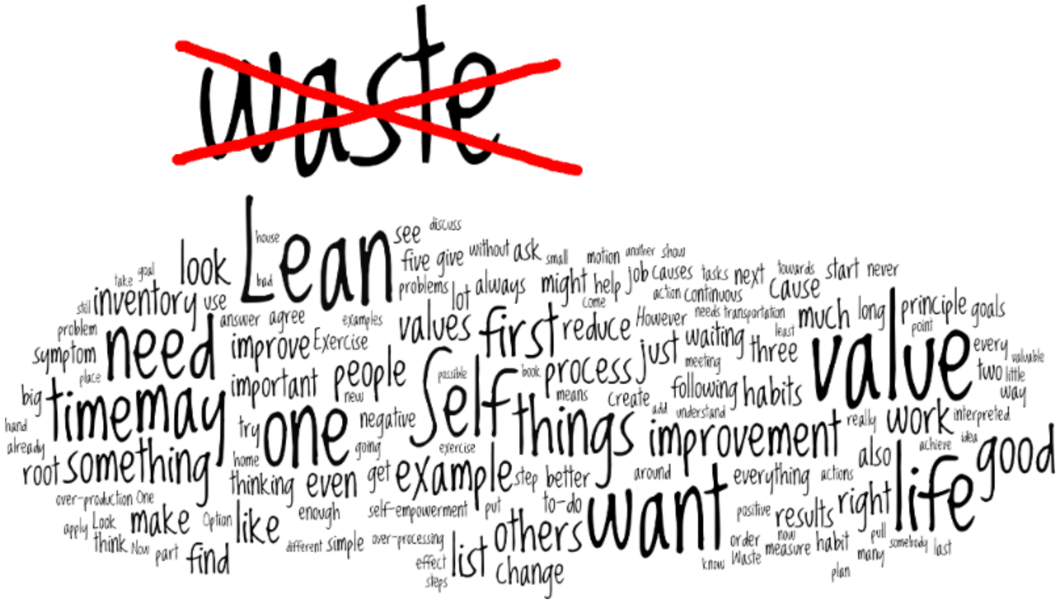
Any Questions about Qualitative Process Analysis?



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